

Microbe

KodaPure · CAT # KBTNA-M-26

IMPORTANT

NOTE BEFORE START

- 1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
- 2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
- 3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
- 4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

| SAMPLES                                                         | SAMPLE INPUT                     | ELUATE STORAGE           |
|-----------------------------------------------------------------|----------------------------------|--------------------------|
| Bacterial/fungal pure culture, dairy, lakewater, internal organ | 1 mL cloudy liquid / ~2 mm solid | 4 °C short / -20 °C long |

REAGENTS SUPPLIED

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| ● 1 X 15 mL LBI Lysis Buffer I       | ● 1 X 37 mL LBII Lysis Buffer II     |
| ● 1 X HB01 Homogenizing Beads 0.1 mm | ● 1 X HB05 Homogenizing Beads 0.5 mm |
| ● 1 X 27 mL BB Binding Buffer        | ● 1 X 0.8 mL MB Magnetic Beads       |
| ● 1 X 37 mL WBIII Wash Buffer III    | ● 1 X 72 mL WBII Wash Buffer II      |
| ● 1 X 5 mL EB Elution Buffer         |                                      |

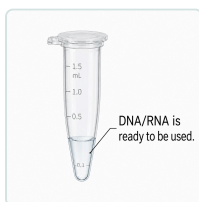
**ADDITIONAL MATERIALS AND APPARATUS REQUIRED**

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- |                                                                                                                      |                                                                                              |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes</li></ul> <hr/> | <ul style="list-style-type: none"><li>• Tube rack</li></ul> <hr/>                            |
| <ul style="list-style-type: none"><li>• Vortex machine or homogenizer</li></ul> <hr/>                                | <ul style="list-style-type: none"><li>• 1.5 mL microcentrifuge tubes</li></ul> <hr/>         |
| <ul style="list-style-type: none"><li>• Magnetic rack</li></ul> <hr/>                                                | <ul style="list-style-type: none"><li>• Pipettes and compatible pipette tips</li></ul> <hr/> |

## METHOD

- Liquid sample:** Transfer 1 mL cloudy liquid to a 1.5 mL microcentrifuge tube. Centrifuge at max speed ( $\sim 12,000 \times g$ ) for 1 min and discard the supernatant. Repeat 2–3 times if the pellet is very small.  
**Solid sample:** Add 1/6 of culture plate or small solid sample ( $\sim 2$  mm diameter or  $\sim 2$  loopfuls) to a 1.5 mL microcentrifuge tube.
- Add homogenizing beads HB01 and HB05 to the sample tube up to  $\sim 5$  mm height. Add 600  $\mu$ L LBI. Then, vortex the tube at 3000–4000 rpm for 3 min.
- Incubate the sample at 56 °C for 30 min. (For fungi/bacterial samples with high protein content, consider adding 5  $\mu$ L of 20 mg/mL Proteinase K before the 56 °C incubation.)
- Centrifuge the tube at max speed for 3–5 min. Then, transfer 250  $\mu$ L of clear supernatant to a new tube. (Avoid taking the dirty pellet.)
- Add 250  $\mu$ L LBI, close the tube lid and mix well. Leave it for 5 min.
- Bring MB to room temperature and mix it well before use. Add 500  $\mu$ L BB (1X total volume) and 15  $\mu$ L MB to the sample tube. Mix it and leave for 5 min.
- Place the tube on the magnetic rack until the supernatant is clear ( $\sim 2$  min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
- Add 700  $\mu$ L WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
- Add 700  $\mu$ L WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 9). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
- Open the tube lid and leave it for 5 min to air-dry.
- Add 50–80  $\mu$ L EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
- The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.



## NOTES

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- DNA-only extraction: (Add RNase A during or after extraction to remove RNA from supernatant):
  - **During extraction:** after Step 4, add 1  $\mu\text{L}$  of 20 mg/mL RNase A, mix, and incubate at 37 °C or room temperature for 30 min. Continue with Step 5.
  - **After extraction:** add 1  $\mu\text{L}$  of 20 mg/mL RNase A to the eluate, mix, and leave at room temperature for 15 min. Your DNA is ready to use.

## TROUBLESHOOTING

| EVENT                    | LIKELY CAUSE                             | SUGGESTED FIX                                                                                                                                                                              |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low DNA yield            | Too much sample, or poor homogenization  | Use less starting material and homogenise fully.                                                                                                                                           |
| RNA in a DNA-only eluate | RNase A not used                         | Add RNase A and incubate as noted.                                                                                                                                                         |
| Bead carryover           | Beads disturbed during transfer          | Leave on the magnet until clear, then transfer carefully.                                                                                                                                  |
| Brown eluate             | Soil, fat, or debris carried over        | Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50. |
| Poor PCR result          | Inhibitor carryover or too much template | Dilute the eluate 1:10 to 1:50 and avoid debris.                                                                                                                                           |
| Soil inhibition          | Humic substances carried over            | Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.                                                                                        |

### SAFETY

### SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.

# Microbe – dirty samples

KodaPure · CAT # KBTNA-MD-26

**IMPORTANT****NOTE BEFORE START**

1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

**PROTOCOL SUMMARY**

| SAMPLES       | SAMPLE INPUT                                | INHIBITOR REMOVAL    | ELUATE STORAGE                      |
|---------------|---------------------------------------------|----------------------|-------------------------------------|
| <b>Faeces</b> | <b>1 mL cloudy liquid /<br/>~2 mm solid</b> | <b>Tube A, 3 min</b> | <b>4 °C short / –20 °C<br/>long</b> |

**REAGENTS SUPPLIED**

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| ● 1 X 15 mL LBI Lysis Buffer I       | ● 1 X 37 mL LBII Lysis Buffer II     |
| ● 1 X HB01 Homogenizing Beads 0.1 mm | ● 1 X HB05 Homogenizing Beads 0.5 mm |
| ● 50 X Tube A Inhibitor-removal tube | ● 1 X 27 mL BB Binding Buffer        |
| ● 1 X 0.8 mL MB Magnetic Beads       | ● 1 X 37 mL WBIII Wash Buffer III    |
| ● 1 X 72 mL WBII Wash Buffer II      | ● 1 X 5 mL EB Elution Buffer         |

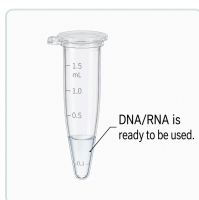
**ADDITIONAL MATERIALS AND APPARATUS REQUIRED**

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- |                                                                                                                      |                                                                                              |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes</li></ul> <hr/> | <ul style="list-style-type: none"><li>• Tube rack</li></ul> <hr/>                            |
| <ul style="list-style-type: none"><li>• Vortex machine or homogenizer</li></ul> <hr/>                                | <ul style="list-style-type: none"><li>• 1.5 mL microcentrifuge tubes</li></ul> <hr/>         |
| <ul style="list-style-type: none"><li>• Magnetic rack</li></ul> <hr/>                                                | <ul style="list-style-type: none"><li>• Pipettes and compatible pipette tips</li></ul> <hr/> |

## METHOD

- Liquid sample:** Transfer 1 mL cloudy liquid to a 1.5 mL microcentrifuge tube. Centrifuge at max speed ( $\sim 12,000 \times g$ ) for 1 min and discard the supernatant. Repeat 2-3 times if the pellet is very small.  
**Solid sample:** Add small solid sample ( $\sim 2$  mm diameter or  $\sim 2$  loopfuls) to a 1.5 mL microcentrifuge tube.
- Add homogenizing beads HB01 and HB05 to the sample tube up to  $\sim 5$  mm height. Add 700  $\mu$ L LBI. Then, vortex the tube at 3000-4000 rpm for 3 min.
- Centrifuge the tube at max speed for 3-5 min. Then, transfer 400  $\mu$ L of clear supernatant to a new tube. (Avoid taking the dirty pellet.)
- Open one Tube A and transfer all Tube A contents to the transferred supernatant. Mix vigorously by hand and leave for 3 min. Centrifuge for another 5 min.
- Transfer 250  $\mu$ L of the treated supernatant into a new tube. Add 250  $\mu$ L LBI, close the tube lid and mix well. Leave it for 5 min.
- Bring MB to room temperature and mix it well before use. Add 500  $\mu$ L BB and 15  $\mu$ L MB to the sample tube. Mix it and leave for 5 min.
- Place the tube on the magnetic rack until the supernatant is clear ( $\sim 2$  min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
- Add 700  $\mu$ L WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
- Add 700  $\mu$ L WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 9). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
- Open the tube lid and leave it for 5 min to air-dry.
- Add 50-80  $\mu$ L EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
- The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.





## NOTES

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- DNA-only extraction: (Add RNase A during or after extraction to remove RNA from supernatant):
  - **During extraction:** after Step 3, add 1  $\mu\text{L}$  of 20 mg/mL RNase A, mix, and incubate at 37 °C or room temperature for 30 min. Continue with Step 4.
  - **After extraction:** add 1  $\mu\text{L}$  of 20 mg/mL RNase A to the eluate, mix, and leave at room temperature for 15 min. Your DNA is ready to use.
- The 'dirty' variant adds Tube A to strip inhibitors from difficult samples before the beads bind the nucleic acid.

## TROUBLESHOOTING

| EVENT                    | LIKELY CAUSE                             | SUGGESTED FIX                                                                                                                                                                              |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low DNA yield            | Too much sample, or poor homogenization  | Use less starting material and homogenise fully.                                                                                                                                           |
| RNA in a DNA-only eluate | RNase A not used                         | Add RNase A and incubate as noted.                                                                                                                                                         |
| Bead carryover           | Beads disturbed during transfer          | Leave on the magnet until clear, then transfer carefully.                                                                                                                                  |
| Brown eluate             | Soil, fat, or debris carried over        | Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50. |
| Poor PCR result          | Inhibitor carryover or too much template | Dilute the eluate 1:10 to 1:50 and avoid debris.                                                                                                                                           |
| Soil inhibition          | Humic substances carried over            | Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.                                                                                        |

### SAFETY

### SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.

# Animal & tissue

KodaPure · CAT # KBTNA-A-26

**IMPORTANT****NOTE BEFORE START**

1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

**PROTOCOL SUMMARY**

## SAMPLES

**Animal tissue,  
meat, organs,  
cultured cells**

## SAMPLE INPUT

**10-20 mg tissue**

## HOMOGENIZATION

**In LBII (see below)**

## ELUATE STORAGE

**4 °C short / –20 °C  
long****REAGENTS SUPPLIED**

● 1 X 27 mL LBII Lysis Buffer II

● 1 X 22 mL BB Binding Buffer

● 1 X 72 mL WBIII Wash Buffer III

● 1 X 5 mL EB Elution Buffer

● 1 X 12 mL LBI Lysis Buffer I

● 1 X 0.8 mL MB Magnetic Beads

● 1 X 72 mL WBII Wash Buffer II

**ADDITIONAL MATERIALS AND APPARATUS REQUIRED**

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|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes</li></ul> <hr/> | <ul style="list-style-type: none"><li>• Pipettes and compatible pipette tips</li></ul> <hr/>                                                                                                                 |
| <ul style="list-style-type: none"><li>• Vortex machine or homogenizer</li></ul> <hr/>                                | <ul style="list-style-type: none"><li>• Proteinase K, 20 mg/mL</li></ul> <hr/>                                                                                                                               |
| <ul style="list-style-type: none"><li>• Magnetic rack</li></ul> <hr/>                                                | <ul style="list-style-type: none"><li>• Heat block or incubator capable of 56 °C</li></ul> <hr/>                                                                                                             |
| <ul style="list-style-type: none"><li>• Tube rack</li></ul> <hr/>                                                    | <ul style="list-style-type: none"><li>• Optional, depending on homogenization method: liquid nitrogen with chilled mortar and pestle, 2.8–4 mm steel or ceramic beads, or homogenizing stick</li></ul> <hr/> |
| <ul style="list-style-type: none"><li>• 1.5 mL microcentrifuge tubes</li></ul> <hr/>                                 |                                                                                                                                                                                                              |

## METHOD

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1. Homogenize or digest the animal sample in LBII using one method below before continuing.

### **Cryogenic grinding**

For high-quality, long-strand DNA: flash-freeze 10–20 mg sample in liquid nitrogen and grind to a fine powder using a chilled mortar and pestle. Transfer the powder immediately to a 1.5 mL tube containing 500 µL LBII. Mix well, add 5–10 µL Proteinase K, and incubate at 56 °C for 30 min.

### **High-speed bead shaking**

For high-throughput processing: add 10–20 mg sample and 2.8–4 mm steel or ceramic beads to a tube containing 500 µL LBII. Vortex at high speed for 3 min until the tissue forms a uniform slurry. Add 5–10 µL Proteinase K and incubate at 56 °C for 30 min.

### **Proteinase K digestion only**

For small tissue amounts: cut 10–15 mg sample into very small pieces. Add 500 µL LBII and 5–10 µL Proteinase K. Incubate at 56 °C for 1–3 h, or until the tissue dissolves. Larger samples may require overnight incubation.

### **Manual mincing or homogenizing stick**

For pretreatment before lysis: add 10–20 mg sample and 500 µL LBII. Grind the sample to a paste-like consistency using a homogenizing stick. Add 5–10 µL Proteinase K and incubate at 56 °C or room temperature for 1–3 h.

2. Centrifuge the tube for 2 min at max speed (~12,000 × g). Then, transfer 200 µL supernatant into a new tube. Avoid the fat layer, pellet, and debris.
3. Add 200 µL LBI, close the tube lid and mix well. Leave it for 5 min.
4. Bring MB to room temperature and mix it well before use. Add 400 µL BB and 15 µL MB to the sample tube. Mix it and leave for 10 min.
5. Place the tube on the magnetic rack until the supernatant is clear (~2 min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
6. Add 700 µL WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack. Repeat this wash if the supernatant is very dirty.
7. Add 700 µL WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step if the supernatant is dirty (Step 7). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
8. Open the tube lid and leave it for 5 min to air-dry.
9. Add 50–80 µL EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.

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10. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.
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11. Salt or inhibitor carryover can disrupt PCR. For species ID, if samples fail to amplify, dilute 1:10 or 1:50 and rerun.



## NOTES

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- DNA-only extraction: (Add RNase A during or after extraction to remove RNA from supernatant):
  - **During extraction:** after Step 2, add 1  $\mu$ L of 20 mg/mL RNase A to the 200  $\mu$ L lysate and incubate at 37 °C or room temperature for 30 min. Continue with Step 3.
  - **After extraction:** add 1  $\mu$ L of 20 mg/mL RNase A to the eluate, mix, and leave at room temperature for 15 min. Your DNA is ready to use.
- Proteinase K (20 mg/mL) is required for animal tissue extraction but is not supplied.
- Choose the homogenization method according to sample toughness, throughput, and downstream DNA-length requirements. Use less starting material if the lysate is thick or difficult to clarify.

TROUBLESHOOTING

| EVENT                    | LIKELY CAUSE                             | SUGGESTED FIX                                                                                                                                                                              |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low DNA yield            | Too much sample, or poor homogenization  | Use less starting material and homogenise fully.                                                                                                                                           |
| RNA in a DNA-only eluate | RNase A not used                         | Add RNase A and incubate as noted.                                                                                                                                                         |
| Bead carryover           | Beads disturbed during transfer          | Leave on the magnet until clear, then transfer carefully.                                                                                                                                  |
| Brown eluate             | Soil, fat, or debris carried over        | Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50. |
| Poor PCR result          | Inhibitor carryover or too much template | Dilute the eluate 1:10 to 1:50 and avoid debris.                                                                                                                                           |
| Soil inhibition          | Humic substances carried over            | Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.                                                                                        |

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.

Soil

KodaPure · CAT # KBTNA-S-26

IMPORTANT

NOTE BEFORE START

- 1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
- 2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
- 3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
- 4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

| SAMPLES                                                              | SAMPLE INPUT | INHIBITOR REMOVAL | ELUATE STORAGE           |
|----------------------------------------------------------------------|--------------|-------------------|--------------------------|
| Soil (Not suitable for soil with extremely high inhibitor/minerals.) | 30 mg soil   | Tube A, 3 min     | 4 °C short / –20 °C long |

REAGENTS SUPPLIED

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| ● 1 X 72 mL PW Pre-Wash Buffer       | ● 1 X 37 mL LBI Lysis Buffer I       |
| ● 1 X HB01 Homogenizing Beads 0.1 mm | ● 1 X HB05 Homogenizing Beads 0.5 mm |
| ● 50 X Tube A Inhibitor-removal tube | ● 1 X 17 mL BB Binding Buffer        |
| ● 1 X 0.8 mL MB Magnetic Beads       | ● 1 X 37 mL WBIII Wash Buffer III    |
| ● 1 X 72 mL WBII Wash Buffer II      | ● 1 X 5 mL EB Elution Buffer         |

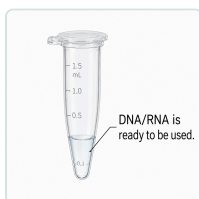


ADDITIONAL MATERIALS AND APPARATUS REQUIRED

- |                                                                                                              |                                                                                      |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Centrifuge machine suitable for 1.5 mL microcentrifuge tubes</li></ul> | <ul style="list-style-type: none"><li>Tube rack</li></ul>                            |
| <ul style="list-style-type: none"><li>Vortex machine or homogenizer</li></ul>                                | <ul style="list-style-type: none"><li>1.5 mL microcentrifuge tubes</li></ul>         |
| <ul style="list-style-type: none"><li>Magnetic rack</li></ul>                                                | <ul style="list-style-type: none"><li>Pipettes and compatible pipette tips</li></ul> |

## METHOD

1. Add 30 mg soil to a 1.5 mL tube.
2. Mix PW well before use. Add 700  $\mu$ L PW and mix vigorously by hand. Leave it for 2 min. Centrifuge at max speed ( $\sim 12,000 \times g$ ) for 2 min. Discard the supernatant and repeat this step.
3. Add homogenizing beads HB01 and HB05 to the sample tube up to  $\sim 5$  mm height. Add 700  $\mu$ L LBI. Then, vortex the tube at 3000–4000 rpm for 5 min.
4. Centrifuge the tube at max speed for 3–5 min. Then, transfer 400  $\mu$ L of clear supernatant to a new tube. (Avoid taking the dirty pellet.) Open one Tube A and add all its contents to the supernatant. Mix vigorously by hand and leave for 3 min. Centrifuge for another 5 min.
5. Transfer 300  $\mu$ L of the treated supernatant into a new tube. Bring MB to room temperature and mix it well before use. Add 300  $\mu$ L BB and 15  $\mu$ L MB to the sample tube. Mix gently and leave for 10 min.
6. Place the tube on the magnetic rack until the supernatant is clear ( $\sim 2$  min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
7. Add 700  $\mu$ L WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
8. Add 700  $\mu$ L WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 8). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
9. Open the tube lid and leave it for 5 min to air-dry.
10. Add 50–80  $\mu$ L EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
11. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.
12. Salt or inhibitor carryover can disrupt PCR; if samples fail to amplify, dilute 1:10 or 1:50 and rerun.



TROUBLESHOOTING

| EVENT                    | LIKELY CAUSE                             | SUGGESTED FIX                                                                                                                                                                              |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low DNA yield            | Too much sample, or poor homogenization  | Use less starting material and homogenise fully.                                                                                                                                           |
| RNA in a DNA-only eluate | RNase A not used                         | Add RNase A and incubate as noted.                                                                                                                                                         |
| Bead carryover           | Beads disturbed during transfer          | Leave on the magnet until clear, then transfer carefully.                                                                                                                                  |
| Brown eluate             | Soil, fat, or debris carried over        | Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50. |
| Poor PCR result          | Inhibitor carryover or too much template | Dilute the eluate 1:10 to 1:50 and avoid debris.                                                                                                                                           |
| Soil inhibition          | Humic substances carried over            | Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.                                                                                        |

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.

Plant

KodaPure · CAT # KBTNA-P-26

IMPORTANT

NOTE BEFORE START

- 1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
- 2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
- 3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
- 4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

| SAMPLES                                        | SAMPLE INPUT | HOMOGENIZATION                | INHIBITOR REMOVAL |
|------------------------------------------------|--------------|-------------------------------|-------------------|
| Leaf, seed, root, stem, and other plant tissue | 50-100 mg    | In LBIII at 65 °C (see below) | Tube A, 3 min     |
| ELUATE STORAGE                                 |              |                               |                   |
| 4 °C short / -20 °C long                       |              |                               |                   |

REAGENTS SUPPLIED

|                                               |                                      |
|-----------------------------------------------|--------------------------------------|
| ● 1 X 12 mL LBI Lysis Buffer I                | ● 1 X 37 mL LBIII Lysis Buffer III   |
| ● 50 X HB045 Homogenizing Beads 0.5 mm & 4 mm | ● 50 X Tube A Inhibitor-removal tube |
| ● 1 X 22 mL BB Binding Buffer                 | ● 1 X 1.1 mL MB Magnetic Beads       |
| ● 1 X 37 mL WBIII Wash Buffer III             | ● 1 X 72 mL WBI Wash Buffer II       |
| ● 1 X 5 mL EB Elution Buffer                  |                                      |

**ADDITIONAL MATERIALS AND APPARATUS REQUIRED**

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- |                                                                                                                      |                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes</li></ul> <hr/> | <ul style="list-style-type: none"><li>• 1.5 mL microcentrifuge tubes</li></ul> <hr/>                                                                                        |
| <ul style="list-style-type: none"><li>• Vortex machine or homogenizer</li></ul> <hr/>                                | <ul style="list-style-type: none"><li>• Pipettes and compatible pipette tips</li></ul> <hr/>                                                                                |
| <ul style="list-style-type: none"><li>• Magnetic rack</li></ul> <hr/>                                                | <ul style="list-style-type: none"><li>• Heat block or incubator capable of 65 °C</li></ul> <hr/>                                                                            |
| <ul style="list-style-type: none"><li>• Tube rack</li></ul> <hr/>                                                    | <ul style="list-style-type: none"><li>• Optional, depending on homogenization method: liquid nitrogen with chilled mortar and pestle, or homogenizing stick</li></ul> <hr/> |

## METHOD

1. Select one homogenization method and process the plant sample in LBIII before continuing.

### Cryogenic grinding

For high-quality, long-strand DNA: flash-freeze 50–100 mg sample in liquid nitrogen and grind to a fine powder using a chilled mortar and pestle. Mix LBIII thoroughly before use. In a 1.5 mL tube, add 700  $\mu$ L pre-warmed LBIII and the grinded sample. Mix well and incubate at 65 °C for 40 min.

### High-speed shaking with homogenizing beads (optional: use KreatBio HB045)

For high-throughput processing: pre-warm LBIII and mix thoroughly. Add 500  $\mu$ L LBIII and one to two 3–5 mm homogenizing beads to a 1.5 mL tube. Add 50 mg sample cut into small pieces, then grind with a homogenizing stick until a slurry forms. If no stick is available, cut the tissue as finely as possible before adding it. Add another 200  $\mu$ L LBIII and mix. Incubate at 65 °C for 15 min, vortex at 3000–4000 rpm for 4 min until uniform, then incubate again at 65 °C for 40 min.

2. Centrifuge the tube at max speed for 5–10 min. Then, transfer 300  $\mu$ L of clear supernatant to a new 1.5 mL tube. (Avoid taking the dirty pellet.) Open one Tube A and transfer all Tube A contents to the transferred supernatant. Mix vigorously by hand and leave for 3 min. Centrifuge for another 5 min.
3. Transfer 200  $\mu$ L of the treated supernatant into a new tube. Add 200  $\mu$ L LBI, close the tube lid and mix well. Leave it for 5 min.
4. Bring MB to room temperature and mix it well before use. Add 400  $\mu$ L BB and 20  $\mu$ L MB to the sample tube. Mix gently and leave for 10 min.
5. Place the tube on the magnetic rack until the supernatant is clear (~2 min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
6. Add 700  $\mu$ L WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack.
7. Add 700  $\mu$ L WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step (Step 7). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
8. Open the tube lid and leave it for 5 min to air-dry.
9. Add 50–80  $\mu$ L EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.
10. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.



## NOTES

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- LBIII with HB045 beads is used to break tough plant cell walls. Pre-warm LBIII and mix it thoroughly before use.
- Choose cryogenic grinding when long-strand DNA is preferred. Choose high-speed shaking with HB045 when processing many samples or when a faster mechanical workflow is preferred.

TROUBLESHOOTING

| EVENT                    | LIKELY CAUSE                             | SUGGESTED FIX                                                                                                                                                                              |
|--------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low DNA yield            | Too much sample, or poor homogenization  | Use less starting material and homogenise fully.                                                                                                                                           |
| RNA in a DNA-only eluate | RNase A not used                         | Add RNase A and incubate as noted.                                                                                                                                                         |
| Bead carryover           | Beads disturbed during transfer          | Leave on the magnet until clear, then transfer carefully.                                                                                                                                  |
| Brown eluate             | Soil, fat, or debris carried over        | Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50. |
| Poor PCR result          | Inhibitor carryover or too much template | Dilute the eluate 1:10 to 1:50 and avoid debris.                                                                                                                                           |
| Soil inhibition          | Humic substances carried over            | Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.                                                                                        |

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.